

ORIGINAL ARTICLE

Specifying the Cardio-Respiratory Patterns During Fast-Paced Breathing

Maša Iskra^{1,2}  | Sylvain Laborde^{1,2}  | Tasha Poppa^{3,4}  | Caterina Salvotti^{1,2}  | Elisa Weinand^{1,5}  | Markus Raab¹  | Laura Voigt¹ 

¹Institute of Psychology, German Sport University Cologne, Cologne, Germany | ²HRV LA Bode—Performance Psychophysiology, German Sport University Cologne, Cologne, Germany | ³International Center for Primate Brain Research, Chinese Academy of Sciences, Shanghai, China | ⁴Laboratory for Neuro- and Psychophysiology, KU Leuven, Leuven, Belgium | ⁵University of Cologne, Cologne, Germany

Correspondence: Maša Iskra (m.iskra@dshs-koeln.de)

Received: 1 December 2025 | **Revised:** 10 April 2026 | **Accepted:** 14 April 2026

Keywords: breathing discomfort | breathing technique | cardio-respiratory coupling | heart rate variability | pre-ejection period

ABSTRACT

Fast-paced breathing (FPB) has emerged as a technique to purposefully increase physiological activation to facilitate motor-cognitive performance. Empirical evidence of prominent physiological processes associated with FPB, cardiac changes however, remain scarce. To address this gap, the present study sought to systematically quantify the changes in heart rate variability and cardiac contractility during FPB at 35 and 55 cycles/min (CPM), compared to spontaneous and paced breathing (6 and 15 CPM), while accounting for breathing discomfort. Healthy, physically active participants ($N = 38$) performed each breathing exercise for 2.5 min, with electrocardiography and impedance cardiography signals recorded throughout. Heart rate variability was indexed through RMSSD, and the pre-ejection period (PEP) served as an index of cardiac contractility. Results showed that FPB at 35 and 55 CPM led to reduced heart rate variability, whereas only the latter condition elicited a significant increase in cardiac contractility, suggesting a more pronounced stimulation. Breathing discomfort level increased as the breathing frequency deviated from the spontaneous range, reaching a moderate level at 55 CPM. These findings elucidate the frequency dependence in reciprocal and independent cardiac patterns, informing conceptual models of paced breathing.

1 | Background

Fast-paced breathing (FPB), characterized by frequencies above 20 cycles/min (CPM), is increasingly applied to up-regulate psychophysiological activation. This process involves deliberate adjustments in physiological activation that support shifts in emotion, cognition, or action to facilitate performance (Nigg 2017). FPB has been shown to reduce reaction and movement times during a motor task (Buchanan and Janelle 2021), increase hand grip strength (Telles et al. 2014), and enhance perceived alertness, energy, and focus (Balters et al. 2018). While research on FPB exercises has predominantly focused on how they affect performance, the accompanying physiological processes remain largely speculative. This mismatch between the current understanding of physiological changes during FPB and the

increasing interest in its practical application provides the rationale for the present study. Building on prior theoretical and empirical work on cardio-respiratory patterns, the study aimed to systematically examine changes in heart rate variability (HRV) and cardiac contractility during FPB exercises in comparison with paced control and slow-paced breathing (SPB).

Early attempts to conceptualize the effects of voluntary breath modification in humans have focused on the bidirectional link between cardio-respiratory and neural circuits, which is anatomically supported via parasympathetic (e.g., vagus) and sympathetic pathways. The functioning of this circuit is illustrated through one of the central cardio-respiratory processes, the respiratory sinus arrhythmia (RSA; Yasuma and Hayano 2004). Respiratory-related changes in HRV

arise from the interaction between neural respiratory and autonomic control centers, eliciting reciprocal parasympathetic withdrawal and sympathetic activation during inspiration and an opposite pattern during expiration (Quigley et al. 2024). Voluntary breath modification is expected to induce alterations in the central autonomic network, which may be quantified through changes in cardiac sympathetic and parasympathetic activity due to structural connectivity (Sevoz-Couche and Laborde 2022; Smith et al. 2017). Two recognized indices are the pre-ejection period (PEP), an index of cardiac contractility and sympathetic beta-adrenergic activity (Cacioppo et al. 1994), and the root mean square of successive differences between heartbeats (RMSSD), an index of vagally-mediated HRV (Laborde et al. 2017; Penttilä et al. 2001). Measurement of these indices could offer an appealing approach to test the reciprocally coupled cardiac sympathetic activation and parasympathetic withdrawal as the proposed pattern accompanying the performance-facilitating effects of FPB (Balters et al. 2018; Malhotra et al. 2022). Nevertheless, the exact interpretation has been met with partially divergent positions.

The interpretation of cardiac patterns as indices of cardiac autonomic activity becomes more nuanced as the breathing pattern deviates from the spontaneous breathing range. This most likely occurs due to dynamic changes in the integrated physiological system, encompassing respiratory and cardiac processes (Ritz 2024). Considering respiratory-induced HRV, two distinct positions evolved. Both positions agree that RSA is predominantly or exclusively vagally mediated and serves an important biological function in meeting behavioral requirements, including the momentary alertness level (Grossman and Taylor 2007; Kromenacker et al. 2018; Lehrer and Gevirtz 2014). However, they diverge about the phasic and tonic vagal implications of changes in RSA.

The phasic position postulates that changes in RSA due to respiratory variations primarily reflect differences in phasic patterning of vagal effects upon heart rate (Grossman and Taylor 2007; Ritz 2024). It stems from findings of pharmacological blockade studies with non-selective β -blockers, in which heart period and RSA did not closely covary with respiration (i.e., 6–20 CPM, Grossman et al. 1991; Grossman and Kollai 1993). The authors interpreted these findings by associating changes in RSA magnitude with phasic vagal influences but not with cardiac vagal tone, indexed through the beta-blocked heart period. According to this position, the quantity of vagal bursts is predicted to be higher during exhalation in SPB than in FPB, while the mean level of vagal discharge remains the same.

The tonic position postulates that voluntary breath modification exerts longer-lasting effects on cardiac autonomic activity via the vagal afferent pathway to brainstem nuclei, and strengthening of the vagal efferent system (Gerritsen and Band 2018; Lehrer and Gevirtz 2014; Noble and Hochman 2019; Shaffer and Meehan 2020). This position points to experimental findings of long-term changes in heartbeat-evoked potentials as a result of SPB training. The changes exceeded those of other relaxation techniques, such as progressive muscle relaxation, and correlated significantly with HRV (Huang et al. 2018). These findings challenge the idea that respiration-induced changes in

HRV are constrained to phasic patterning and align with meta-analytic findings on immediate and long-term effects of voluntary slow breathing (Laborde et al. 2022a).

The debate on peripheral and central processes accompanying changes in HRV is ongoing (Quigley et al. 2024), with insufficient empirical evidence to side with either existing theoretical position. Even recently, the relationship between RSA and cardiac vagal activity continues to be described with caution: “Changes in respiratory parameters will, in turn, cause alterations in RSA magnitude that may or may not be related to variations in cardiac vagal tone” (Grossman 2023, 5). Distinguishing these two positions would necessitate selective pharmacological β -blockade or invasive measures under varying breathing patterns, beyond the scope of the present study. Given the implementation of FPB exercises centers around immediate psychophysiological changes rather than adaptations over time, it does not exclusively depend on one interpretation and may be achieved through (a) changes in the phasic patterning of vagal influences upon the heart and/or (b) alterations in overall cardiac vagal activity. We will henceforth refer to the specific cardiac parameters in our predictions and discuss the findings using a balanced approach.

Empirical investigations of cardiac patterns during paced breathing have primarily examined heart rate and HRV. The results show opposing patterns: robust increases in HRV during SPB at 6 CPM, and sparsely reported decreases during FPB, with a corresponding elevation in heart rate, systolic, and diastolic blood pressure (Buchanan and Janelle 2021; Nayak et al. 2015; Peng et al. 2004). No effect of paced breathing on PEP has been found in the range of 5–25 CPM (Houtveen et al. 2005). Considering that FPB frequencies barely exceeded the spontaneous range, and that PEP measurement may have been affected by methodological limitations arising from postural changes, this relationship necessitates further investigation. Beyond differences at the mean level, dynamic changes in cardio-respiratory interactions are increasingly attracting research interest. During SPB at 6 CPM, the lag between respiratory and cardiac oscillations decreases, resulting in strong cardio-respiratory coupling with a 0° phase relationship (Lehrer and Gevirtz 2014). This well-established pattern makes SPB at 6 CPM a suitable comparison condition to FPB, for which the changes in cardio-respiratory coupling remain unclear. These converging research gaps call for a systematic approach to testing cardio-respiratory patterns across breathing frequencies.

One of the key methodological challenges in FPB protocols is high heterogeneity. The frequencies of existing exercises range from 25 to 145 CPM (e.g., Buchanan and Janelle 2021; Joshi and Telles 2009; Peng et al. 2004), with no population- or purpose-specific recommendations. In sedentary individuals, FPB has been implemented at 25–35 CPM (Buchanan and Janelle 2021; Denot-Ledunois et al. 1998), whereas higher frequencies are found in physically active samples (30–60 CPM; Nalbandian et al. 2017; Sonetti et al. 2001). The marked heterogeneity of FPB exercises across applied and research fields echoes the lack of consensus and guidelines, the development of which depends on a clearer understanding of the physiological basis as a first step.

The present study sought to experimentally test the changes in cardiac patterns during and after FPB (at 35 and 55 CPM)

in comparison to a paced control condition at 15 CPM and SPB at 6 CPM. Two FPB frequencies were selected to simulate a milder and more pronounced stimulation. Reciprocally coupled decreases in RMSSD and PEP are considered facilitative for up-regulating physiological activation (Fisher et al. 2015). Subjective breathing discomfort was measured to assess how well participants were able to accommodate the deviation from their spontaneous respiratory frequency, reflecting the feasibility of the protocol.

Pre-registered hypotheses for between-condition comparisons were derived from prior findings indicating more pronounced reciprocal coupling between parasympathetic withdrawal and sympathetic activation with increasing physical and cognitive demands (Berntson et al. 1994; Fisher et al. 2015; Lenneman and Backs 2009). Lowest PEP and RMSSD values were expected at 55 CPM, whereas the highest PEP and RMSSD values were expected at 6 CPM. Last, in line with Denot-Ledunois et al. (1998), we hypothesized that the breathing discomfort would increase with a deviation in breathing frequency from the spontaneous range, resulting in the highest breathing discomfort at 55 CPM.

Due to the scarcity of research on the physiological changes during FPB at different frequencies, we tentatively predicted a sequential, frequency-dependent response by referring to changes in cardiac responses to different exercise intensities (Fisher et al. 2015). A decrease in RMSSD would thereby occur first (i.e., at 35 CPM), while an increase in PEP would tend to occur only at higher frequencies (i.e., at 55 CPM), although the cut-off point could not be pre-determined. Duration of cardiac effects and coupling between cardiac and respiratory signals were addressed in exploratory analyses.

2 | Methods

2.1 | Participants

Physically active individuals, engaging in sport activities for at least 2 h per week (Klaperski et al. 2013), aged 18 to 30 years, were recruited using social media networks at a local university. The sample selection reflects the aim to develop breathing exercises specifically applicable in younger adults within performance settings. The required sample size was determined using an a priori power analysis conducted in G*Power 3.1 (Faul et al. 2009). The expected effect size was based on the lower bound of the 95% confidence interval for the effect size reported by Buchanan and Janelle (2021) in relation to the effects of FPB on RMSSD. This study was used due to the similarity of our study design. Given an effect size $f=0.226$, for repeated-measures ANOVA, with statistical power set at 0.95, an α -level of 0.05, the correlation among repeated measures $r=0.50$, and five measurements, the minimum required sample size was determined to be $N=38$. The exclusion criteria comprised medication intake, self-reported cardiovascular and neurological conditions, asthma, and diabetes (Laborde et al. 2017). In participant recruitment, we accounted for previously reported high exclusion rates in ICG analyses due to movement artifacts (Sheikh et al. 2020). The sample consisted of 58 physically active participants (23 female; $M_{\text{age}} = 23.0$, $SD = 2.4$; number of sport participation h/

week: $M = 9.0$, $SD = 4.5$). Twelve participants reported having practiced breathing techniques. The experimental protocol was approved by the Ethics Committee of the local university (Project Identification Code 122/2023).

2.2 | Breathing Manipulation

2.2.1 | Pilot Study

To address the large heterogeneity in FPB exercises and make an informed selection of breathing frequencies, we first conducted a pilot study, using a within-subject design with seven conditions (Supplement S1): SPB6, PB15, FPB25, FPB35, FPB45, and FPB55. Upon analyzing the changes in RMSSD, the number of conditions for the main experiment was limited to five, including 35 and 55 CPM as the two selected FPB frequencies. This allowed a gradual change in frequency of 20 CPM, enabled a more complete randomization and counterbalancing of the conditions, and reduced the time participants were required to refrain from spontaneous breathing.

2.2.2 | Main Experiment

Participants were instructed to breathe at 6, 15, 35, and 55 CPM. They followed a breathing pacer on a computer screen, created in RStudio (version 2023.06.1; R Core Team 2022) and PsychoPy (version 2022.2.4; Peirce et al. 2022). The breathing pattern was portrayed as a ball moving upwards and downwards to indicate inspiration and expiration, respectively (cf. Laborde et al. 2022b).

The common baseline frequency found in previous literature was 15 CPM (e.g., Buchanan and Janelle 2021; Busch et al. 2012; Pinna et al. 2006) and was therefore used as a paced breathing (PB) control within the range of spontaneous breathing. This control condition accounted for the potential confounding effects of paced breathing on cardiac parameters. Consistent with our pilot testing and previous recommendations (Buchanan and Janelle 2021), a specific interval (i.e., 20 CPM) was chosen to differentiate PB from FPB conditions.

To distinguish FPB from hyperventilation, a technique that can be induced at a normal or fast respiratory rate by increasing respiratory depth (McArdle et al. 2015), participants were instructed to breathe at a shallow depth. This was done to avoid undesirable effects such as dizziness, panic, and agitation, which may arise due to the reduction in partial pressure of carbon dioxide and an increase in pH (Gilbert 1999; Ley 1994). The inhalation and exhalation followed a 1:1 ratio in all conditions to avoid the confounding effect of unequal phases (Bae et al. 2021). While previous studies implemented one to five-minute intervals of FPB (e.g., Bellissimo et al. 2020; Pradhan 2013; Sharma et al. 2014), breathing conditions in the present study were performed for 2.5 min to account for an adjustment period to the new breathing rate (see Herrero et al. 2018), while ensuring a reliable assessment of HRV (Esco and Flatt 2014). Breathing depth was measured using the respiratory belt to qualitatively assess whether participants maintained constant depth throughout each condition.

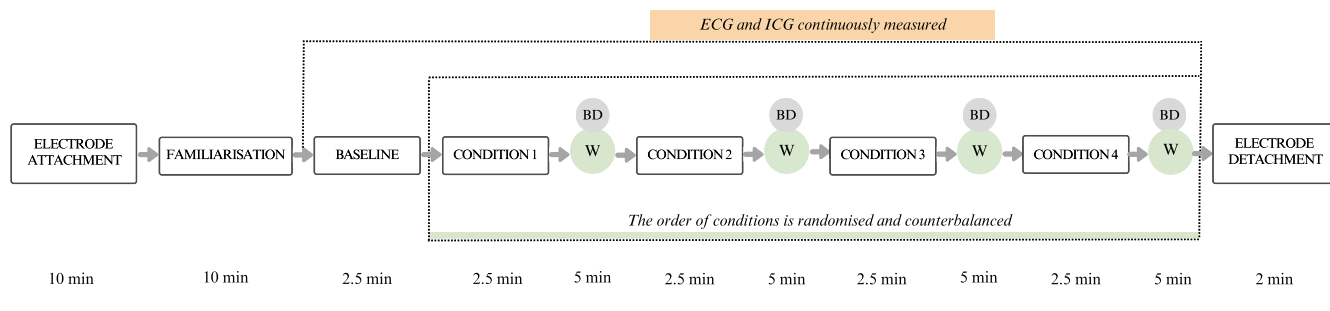


FIGURE 1 | Study protocol. W = washout period; BD = breathing discomfort, conditions = SPB6, PB15, FPB35, FPB55.

2.3 | Experimental Design and Procedure

A within-subject design with five conditions (spontaneous breathing, SPB6, PB15, FPB35, and FPB55) was implemented. Participants attended a single laboratory session for 70 min. They received instruction on following their usual sleep routine, avoiding intense physical activity and alcohol intake, as well as refraining from eating and drinking for 2 h before the experiment (Laborde et al. 2017; Quigley et al. 2024). Participants were asked to complete the informed consent form and a demographic questionnaire, which included items on variables that could influence HRV. The experimental protocol (Figure 1) began with the attachment of the electrodes and the respiratory effort transducer. After skin scrubbing using abrasive gel, the participants familiarized themselves with the correct execution of the paced breathing exercises using video instructions. All participants began with the baseline measure during which they breathed at their spontaneous pace. The order of the following four paced breathing conditions was block randomized and counterbalanced. Each condition was performed for 2.5 min, followed by a five-minute washout period (Gholamrezaei et al. 2021), during which participants evaluated their breathing discomfort. During the experiment, participants were seated with knees at a 90° angle (Laborde et al. 2017). At the end of the experiment, the electrodes were detached, and the participants were debriefed.

2.4 | Physiological Outcomes

The electrocardiography (ECG) and impedance cardiography (ICG) signals were obtained using a BIOPAC MP36R device and AcqKnowledge software (Biopac Systems Inc. 2022) with a sampling frequency of 2000 Hz. RMSSD was selected as an index of HRV and PEP as an index of cardiac contractility (Cacioppo et al. 1994; Laborde et al. 2017; Penttilä et al. 2001; Quigley et al. 2024). To test the hypotheses, ECG and ICG data collected during baseline and in each breathing condition were analyzed.

2.4.1 | Electrocardiography (ECG)

For ECG, a 2-lead setup with pre-gelled Ag/AgCl spot electrodes was used in a Lead-II configuration with the ground connection from the ICG to avoid ground loops. RMSSD was chosen among the HRV parameters because of its physiological origin and relative robustness to respiratory influences, such as fluctuations in breathing patterns (Hill et al. 2009; Laborde et al. 2017; Penttilä et al. 2001). We bandpass-filtered the ECG signal (0.5–50 Hz).

The QRS complex was located using an automated ECG complex boundary function (Biopac Systems Inc. 2022). Detected R-points and Q-points were visually inspected and corrected, if necessary, by adjusting the marker's position. RMSSD values were extracted from the last 2-min period of ECG recordings in each condition. A 2-min recording is deemed sufficient to assess the time domain components of HRV with good reliability (Esco and Flatt 2014). Higher RMSSD values are interpreted as increased HRV. Heart period values (in ms) were additionally extracted to facilitate the interpretation. During FPB, breathing rate may exceed half of the heart rate, which could result in cardiac aliasing, that is, a distortion of the apparent heart period pattern due to undersampling of the cardiac signal relative to respiration (Gersten et al. 1999). Although the impact of this sampling issue on time-domain indices such as RMSSD remains unclear, we provide a magnitude estimate of aliasing at 35 and 55 CPM in Supplement S2.

2.4.2 | Impedance Cardiography (ICG)

For ICG, eight Ag/AgCl pre-gelled, disposable spot electrodes were attached at the base of the neck and on the thorax at the level of the xiphoid process on both sides of the body. PEP is defined as the interval between Q-onset (defined from the ECG) and B-point (defined from the first derivative of the ICG). Rule-based algorithms for identifying the B-point often fall short due to substantial inter- and intra-individual variability in dZ/dt waveform morphology. As a result, these approaches struggle to generalize across subjects or even within the same individual across time. We developed a novel ensemble averaging technique for PEP estimation that was robust to morphological variations (Figure 2).

First, the ICG dZ/dt signal was band-pass filtered between 1 and 35 Hz using a finite impulse response (FIR) filter. The ECG was high-pass filtered at 1 Hz to remove potential baseline drift, and R-wave timing was extracted to define epochs for each beat. A three-stage noise removal algorithm was adapted from an open-source toolbox for ICG analysis (Sheikh et al. 2020) to identify and remove noisy epochs from the recordings. A custom routine for ensemble averaging was subsequently applied: the synchronized ECG and ICG epochs were averaged (thus defining EA-ECG and EA-ICG), then the Q- and S-waves were identified from the EA-ECG as the minima within a window relative to the R-wave. The C-point was identified as the maxima of the EA-ICG. The B-point was identified as the maximum of the inverse gradient of the point-wise standard deviation of the EA-ICG signal within adaptive temporal bounds that were defined

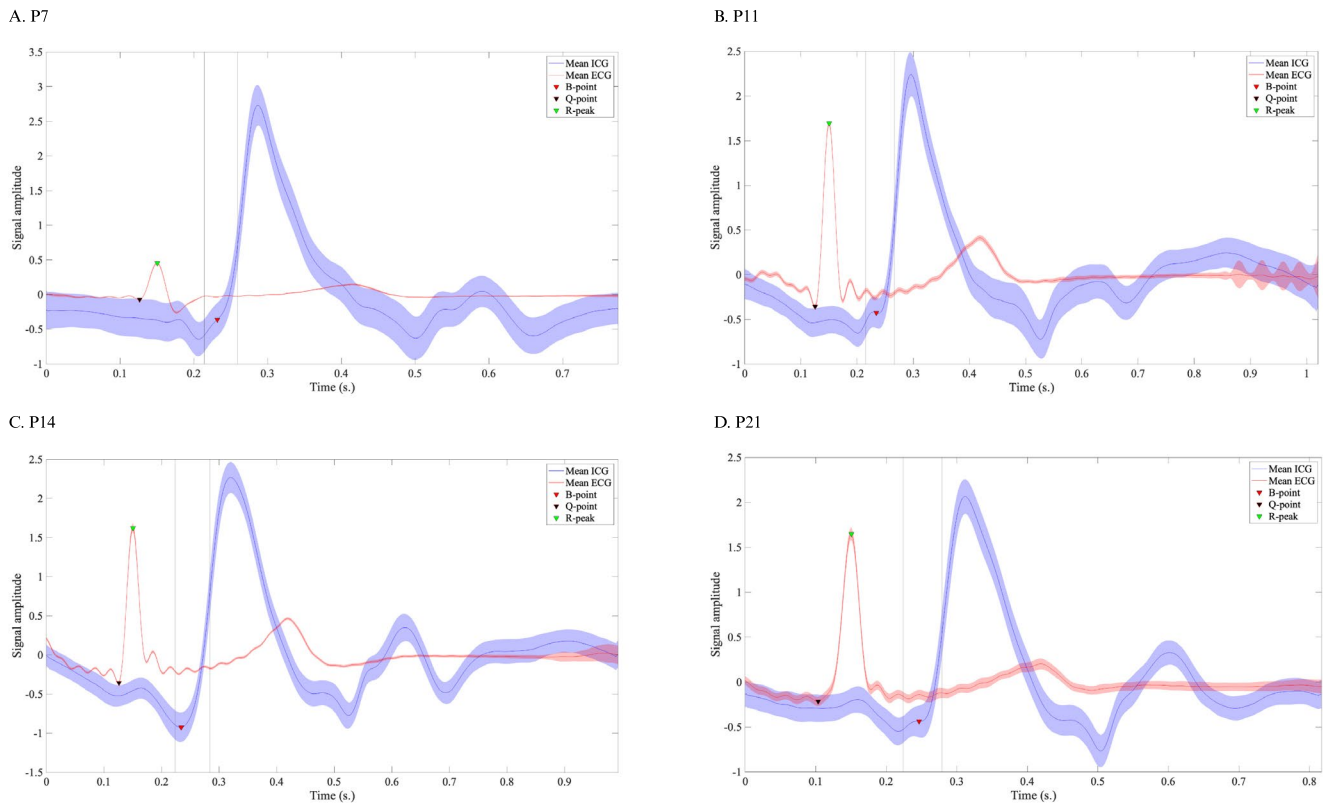


FIGURE 2 | The application of the ensemble average technique across selected participants with distinct dZ/dt waveform morphologies during FPB55. Examples of ensemble-averaged ECG (red) and ICG (dZ/dt , blue) signals from several representative study participants. ICG exhibits strong inter-and-intra subject morphological variability that can make reliable B-point detection a challenge. The red inverted triangles reflect the flexibility of the novel B-point detection algorithm used in this study to robustly identify B-points across variable morphologies. Árbol et al. (2017) qualitatively characterized these features as a change in the gradient (A), a notch (B), an onset of the slope rise (C), and a plateau (D).

from the S-wave and C-point. PEPq values were extracted from the ensemble-averaged signal for the last two-minute period of each condition. Shorter PEP is considered a marker of increased myocardial contractility and beta-adrenergic activity, generally ranging 70–170 ms in resting conditions (Árbol et al. 2017). Assessment of PEP is sensitive to mechanical influences (Newlin and Levenson 1979). Deeper breathing may amplify the oscillations in cardiac preload and afterload via changes in intra-thoracic pressure, producing greater PEP variability during SPB than FPB (Lalande et al. 2012). Although these mechanical influences are largely specific to the respiratory phase (Karlocai et al. 1998) and can be attenuated through abdominal rather than thoracic breathing, the shortening of PEP during SPB should be interpreted cautiously.

2.4.3 | Phase Locking Value

Heart rate and respiration exhibit apparent time-varying phase coherence. Under SPB6, heart rate and respiration can exhibit (near) perfect phase coherence and phase differences around 0° . However, there are likely substantial individual differences in phase coherence across breathing rates. To explore the question of cardio-respiratory coherence across breathing conditions in the present study, including breathing rates at which coherence breaks down, we computed a metric known as the phase locking value (PLV; Lachaux et al. 1999). PLV was used to quantify the

consistency of phase synchronization between respiration and instantaneous heart rate.

$$PLV = \left| \left(\frac{1}{N} \right) \sum_{n=1}^N e^{i(\varphi_{HR}(n) - \varphi_{Resp}(n))} \right|$$

A constant phase difference across cycles yields a PLV near 1, reflecting strong phase synchrony. Conversely, if phase differences vary randomly from cycle to cycle, the PLV approaches 0. However, PLV does not convey the direction or magnitude of the lead-lag relationship between signals; it only reflects the consistency of their phase difference. Importantly, PLV is only valid for signals that oscillate at the same frequency. As such, PLV is expected to be low during the FPB conditions. We also characterized lead-lag relationships by computing the mean, standard deviation, and variance of the phase angle differences using the formulas provided in the CircStat toolbox (Berens 2009).

2.4.4 | Respiratory Depth

Abdominal movement (i.e., changes in the circumference) during breathing exercises was recorded using the respiratory transducer (TSD201; Biopac Systems Inc. 2022). The placement of the strap around the abdomen at the level of the navel reflected the focus of breathing exercises on abdominal rather than thoracic breathing. The signal amplitude was quantified

TABLE 1 | Characteristics of the breathing conditions and descriptive statistics for RMSSD, PEP, heart period, and breathing discomfort ($M \pm SD$).

Condition	Respiratory frequency [CPM]	Respiratory depth [mV]	RMSSD [ms]	PEP [ms]	Heart period [ms]	Breathing discomfort
Baseline	15.04 ± 3.45	0.27 ± 0.43	51.69 ± 24.50	116.96 ± 11.43	901.23 ± 110.90	/
SPB6	6.13 ± 0.17	0.64 ± 0.96	75.97 ± 28.48	114.46 ± 11.16	925.72 ± 107.32	2.54 ± 2.46
PB15	15.39 ± 0.91	0.38 ± 0.49	49.95 ± 25.37	116.17 ± 11.75	904.42 ± 119.94	1.38 ± 1.44
FPB35	35.19 ± 0.83	0.24 ± 0.28	39.39 ± 18.55	115.59 ± 10.87	912.66 ± 126.02	3.56 ± 2.46
FPB55	56.61 ± 1.88	0.18 ± 0.24	24.24 ± 8.28	114.26 ± 10.25	904.34 ± 114.95	5.09 ± 2.96

Note: PEP, pre-ejection period; RMSSD, root mean square of successive differences.

using Acqknowledge software to estimate the respiratory depth in mV (Houtveen et al. 2005).

2.5 | Subjective Outcome

The breathing discomfort level was assessed using a 10-point scale, ranging from no discomfort at all (0) to maximal breathing discomfort (10). This scale aligns with the overall perceived breathing discomfort scale (A1) of the Multidimensional Dyspnea Profile, which is deemed sufficient for laboratory research and was found to highly correlate with the commonly used modified Borg scale (Banzett et al. 2015). The choice of the breathing discomfort scale deviated from the pre-registration to ensure equal intervals between values, enabling the use of parametric tests and improving interpretability.

2.6 | Data Analysis

Data were checked for outliers and potential violations of normality assumptions. As RMSSD data were non-normally distributed, log transformation was applied (Laborde et al. 2017). As part of the manipulation check, adherence to specified breathing paces was visually inspected, and substantial deviations from the set paces were reported. If data segments required for the analysis involved movement artifacts that could not be corrected or the corrected peaks exceeded 5% of all the peaks within an individual condition, these segments were considered as missing data. A participant was excluded from the analysis if they had missing data during baseline or any of the breathing conditions. To decrease the influence of outliers without removing them, five values were winsorized to within three standard deviations of the mean (e.g., Lent et al. 2024). Three participants were excluded from the main analysis due to technical issues with the breathing pattern recording, incomppliance with the instructed breathing pattern, and physiological artifacts in the ECG signal. An additional 17 participants were excluded from the analysis of PEP due to movement artifacts in the ICG signal, yielding 38 final datasets. Finally, for the exploratory analyses of washout periods ($n=31$) and cardio-respiratory phase coupling ($n=32$), additional datasets had to be excluded due to movement artifacts at the start point of the washout period and the sensitivity of the analysis to an inconsistent breathing pattern.

To test the hypotheses, three separate univariate repeated-measures ANOVAs with Greenhouse–Geisser correction were conducted, with breathing conditions as independent variables, and RMSSD, PEP, and breathing discomfort level as dependent variables. In case of a significant main effect, we followed up with pairwise comparisons and applied a Bonferroni correction. The analyses and data visualization were conducted using the JASP software (version 0.18.3; JASP Team 2024) and RStudio (version 2023.06.1; R Core Team 2022), with statistical significance level set at $\alpha=0.05$ throughout.

The exploratory analyses of washout periods were conducted using repeated-measures ANOVAs to examine changes in RMSSD and PEP during the 2min following conditions that elicited significant effects. The correlation between RMSSD and PEP values at baseline and across conditions, as well as the correlation between breathing discomfort and cardiac indices, was quantified using Spearman's rho test due to the violation of the pairwise normality assumption. For cardio-respiratory coupling, statistical significance was assessed by generating a surrogate (null) distribution of PLVs for each breathing condition and participant. This was achieved by computing PLVs using mismatched respiratory phase data from all other participants, thereby disrupting any true physiological coupling. PLVs for the correctly matched data pairs were then computed. A subject's matched PLV was deemed statistically significant if it exceeded the 95th percentile of their respective surrogate distribution (cf. Choe et al. 2021). A Cochran's Q test was used to analyze the binary outcome (above/below significance threshold) across the five conditions, with the McNemar test and Greenhouse–Geisser correction for post hoc tests.

3 | Results

The quantification of respiratory frequency during paced breathing conditions indicated that participants successfully followed the pacer and achieved targeted frequencies. The breathing pattern was additionally characterized in relation to respiratory depth, which decreased as the frequency increased. The highest depth was recorded during SPB6, increasing more than two-fold from the baseline. The descriptive statistics, including mean RMSSD, PEP, heart period, and breathing discomfort values for each condition, are provided in Table 1.

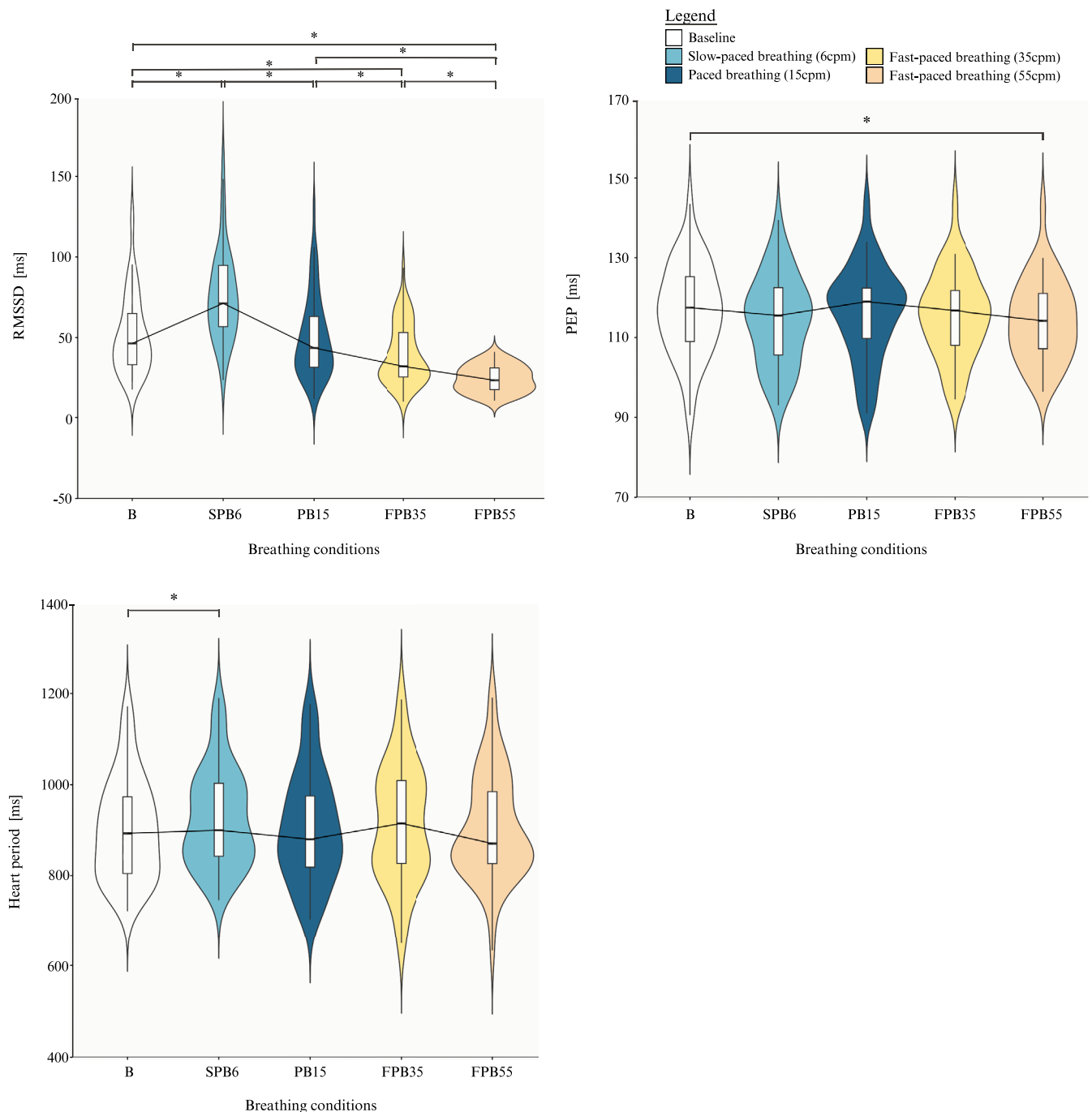


FIGURE 3 | The changes in RMSSD, PEP, and heart period during different breathing conditions. Error bars represent the standard deviation.

3.1 | Cardiac Changes During Paced Breathing

Changes in RMSSD, PEP, and heart period during each breathing condition are visualized in Figure 3. For logRMSSD, a main effect of breathing condition was found, $F(4, 216)=123.93$, $p<0.001$, $\eta_p^2=0.70$. Post hoc t -tests with Bonferroni correction showed that during SPB6, logRMSSD values were significantly higher compared to PB15, $t(54)=8.69$, $p<0.001$, $d=1.07$, 95% CI [0.62, 1.53], indicating higher HRV. LogRMSSD values during FPB55 were significantly lower compared to PB15, $t(54)=12.90$, $p<0.001$, $d=1.52$, 95% CI [0.97, 2.06]. Likewise, lower HRV was recorded during FPB35 compared to PB15, $t(54)=4.68$,

$p<0.001$, $d=0.51$, 95% CI [0.14, 0.88]. LogRMSSD values during FPB35 were higher than during FPB55, $t(54)=8.84$, $p<0.001$, $d=1.01$, 95% CI [0.56, 1.45], suggesting that the interval of 20 CPM led to significantly different levels of HRV. No significant differences between baseline and PB15 were found ($p=0.353$).

For PEP, a main effect of breathing condition was found, $F(4,148)=3.23$, $p=0.014$, $\eta_p^2=0.08$. Post hoc t -tests with Bonferroni correction showed that PEP values during FPB55 were significantly lower compared to baseline, $t(37)=3.01$, $p=0.031$, $d=0.24$, 95% CI [-0.01, 0.50], indicating increased cardiac contractility. Other comparisons did not yield significant results.

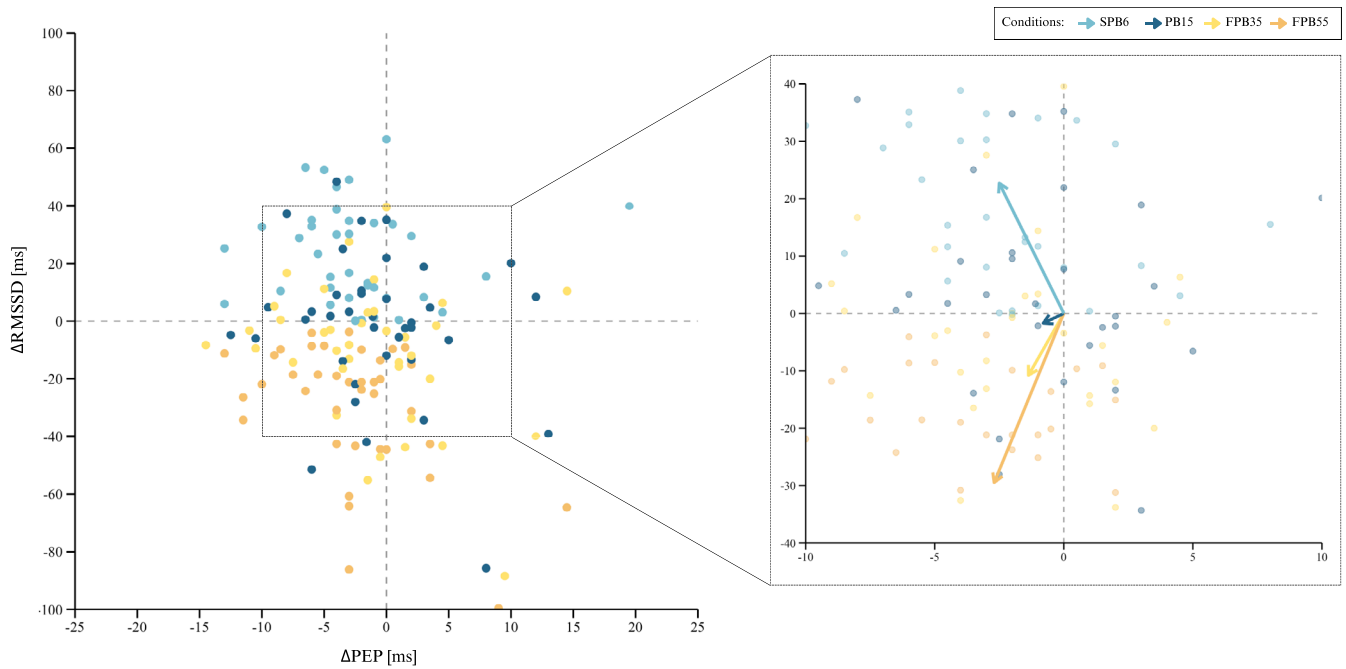


FIGURE 4 | Mapping the relative cardiac responses across the breathing conditions. Cardiac activity changes from baseline are portrayed for each participant and computed across all participants using vectors in a zoomed-in area. Each arrow vector represents the mean response for a specific breathing condition.

For the heart period, a main effect of breathing condition was found, $F(3.11, 161.77) = 3.23$, $p = 0.023$, $\eta_p^2 = 0.06$. Post hoc t -tests with Bonferroni correction showed that during SPB6, mean heart period was significantly longer compared to baseline, $t(52) = 3.14$, $p = 0.02$, $d = 0.21$, 95% CI [0.01, 0.42], indicating slower heart rate. The lowest median heart period value was recorded during FPB55 ($Mdn = 870$), followed by PB15 ($Mdn = 880$) and baseline ($Mdn = 893$). This was additionally reflected by the density of the violin plots (Figure 3). No significant correlations between breathing depth and cardiac activity (RMSSD, PEP, heart period) were found.

The combined assessment of RMSSD and PEP enables us to map the changes across the breathing conditions (cf. Berntson et al. 1994, Figure 4). The lower left quadrant represents a reciprocal activation response, characterized by decreased HRV and increased cardiac contractility. While the mean vectors of PB15, FPB35, and FPB55 are all positioned in this quadrant, statistically significant changes for both measures were only found for FPB55, suggesting this condition elicited reciprocal activation. The upper left quadrant, where the mean vector of SPB6 is positioned, represents a coactivation pattern, although it should be noted that only the change in HRV during SPB6 was statistically significant. Despite the mean reciprocal/coactivated patterns of activity, no significant correlations between RMSSD and PEP were found at baseline ($r_s = -0.19$, $p = 0.263$) or any of the conditions. The correlations for most conditions were close to zero, suggesting independence of the two variables.

3.2 | Breathing Discomfort

For breathing discomfort, a main effect of breathing condition was found, $F(2.33, 125.67) = 43.65$, $p < 0.001$, $\eta_p^2 = 0.45$.

Breathing discomfort scores during PB15 were significantly lower compared to FPB35, $t(54) = 7.21$, $p < 0.001$, $d = 0.91$, 95% CI [0.46, 1.35], and FPB55, $t(54) = 9.71$, $p < 0.001$, $d = 1.55$, 95% CI [1.00, 2.10]. A decrease in frequency to 6 CPM likewise led to a significant increase in breathing discomfort, $t(54) = 3.87$, $p < 0.001$, $d = 0.48$, 95% CI [0.09, 0.88]. These results indicate that participants experienced more discomfort as the breathing frequency deviated from the spontaneous range, reaching the highest discomfort level during FPB55. No significant correlations between breathing discomfort and cardiac activity (RMSSD and PEP) were found for any of the conditions.

3.3 | Exploratory Analyses

3.3.1 | Washout Period

First, the average respiratory frequency during the 2 min succeeding each breathing condition was calculated. The lowest value was recorded after PB15 ($M = 13.38 \pm 3.10$), followed closely by SPB6 ($M = 13.58 \pm 3.62$) and FPB35 ($M = 14.40 \pm 3.94$). The baseline frequency was only surpassed during the washout period after FPB55 ($M = 15.75 \pm 4.54$). The highest proportion of participants breathing below the spontaneous range (< 12 CPM) was recorded after SPB6 (34.69%), whereas the upper threshold of the spontaneous range (> 20 CPM) was surpassed most often after FPB55 (12.24%).

Second, exploratory analysis of HRV during washout periods was conducted using a 1×4 (conditions: baseline, W6, W35, W55) rmANOVA. No main effect of the breathing condition was found on logRMSSD, $F(2.61, 135.50) = 1.00$, $p = 0.388$, suggesting that HRV values returned to baseline within 2 min after the cessation of all breathing conditions. Cardiac contractility during washout was compared to baseline exclusively

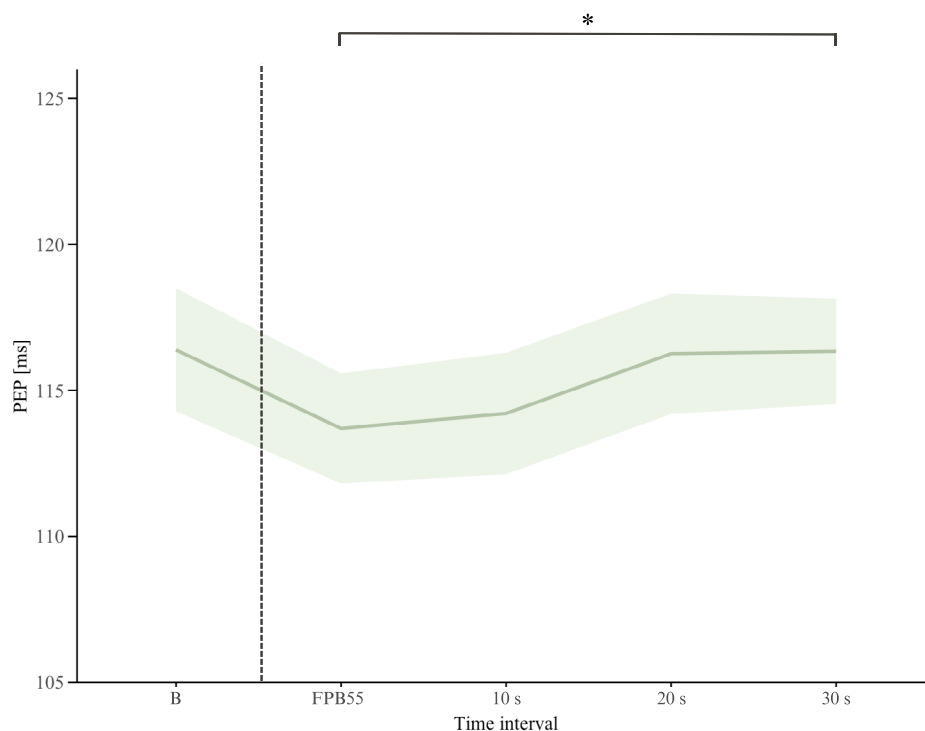


FIGURE 5 | Changes in PEP over 30s after FPB55. Error band represents the standard error.

for FPB55, as the only condition that elicited significant changes. For the 31 datasets for which washout PEP values could be accurately computed, the 2-min averaged PEP value ($M=116.42 \pm 11.46$) closely approximated the baseline PEP ($M=116.39 \pm 11.70$). Using 10-s epochs (cf. Czarnek et al. 2021), we additionally computed the initial changes in PEP over the first 30 s (Figure 5). A 1×4 (conditions: FPB55, 10 s, 20 s, 30 s) rmANOVA showed a significant effect of the timepoint on PEP, $F(2.39, 71.76)=5.34$, $p=0.004$, $\eta_p^2=0.15$. The post hoc t -test indicated that PEP was significantly higher during the third washout interval (i.e., 20–30 s) compared to FPB55, $t(30)=3.76$, $p=0.004$, $d=0.24$, 95% CI [0.01, 0.48], whereas other comparisons did not yield significant results. The PEP values and breathing rate during the washout period were not correlated. This suggests that cardiac contractility gradually decreased after FPB55 and reached a significantly lower level 20–30 s after the cessation of the paced breathing condition.

3.3.2 | Cardio-Respiratory Coupling

PLVs decreased with increasing breathing frequency. The greatest PLVs were observed during SPB6, followed by baseline and PB15, for which similar PLVs were obtained. An exemplary visualization of the coupling between heart rate and respiratory signals, as well as average PLVs across subjects, are provided in Supplement S3. Mean phase angle differences between the respiration and heart rate were computed only for baseline, SPB6, and PB15, given the low number of datasets above the chance threshold for the FPB conditions. The mean above-chance PLV during SPB6 was 0.83, indicating strong phase coherence characterized by a phase angle difference of around 4° (heart rate

leading). The negative phase angle differences obtained during baseline indicated that the heart rate signal lagged behind respiration. The lag further increased during PB15.

Cochran's Q test showed a statistically significant difference in the proportion of datasets that surpassed the significance threshold across the breathing conditions, $\chi^2(4)=179.98$, $p<0.001$. Post hoc McNemar's tests with Bonferroni correction indicated a significantly higher proportion in SPB6 compared to PB15 ($p=0.016$) but not compared to baseline. The proportion of datasets surpassing the threshold in both FPB conditions was significantly lower compared to baseline ($p<0.001$) and PB15 ($p=0.012$ for FPB35, $p=0.001$ for FPB55). This suggests that the observed heart rate–respiration phase locking during FPB was more likely to reflect chance-level coupling, compared to stronger coupling during baseline. Additionally, a significant difference was found between the baseline and PB15, with a higher proportion of datasets in the former condition ($p=0.039$).

4 | Discussion

The present study sought to systematically quantify the changes in RMSSD and PEP during paced breathing at 6, 15, 35, and 55 CPM. Physiological changes related to the cardio-respiratory coupling and the duration of elicited effects were further explored. Subjective experience of breathing discomfort was evaluated to infer feasibility. We tested whether there is evidence for a reciprocally coupled decrease in HRV and an increase in cardiac contractility during FPB. A reciprocally coupled response was identified during 55 CPM, whereas only a significant HRV reduction was elicited during 35 CPM.

4.1 | Changes in Cardiac Activity During Paced Breathing

4.1.1 | Comparison Conditions

Spontaneous breathing at baseline, PB15, and SPB6 served as comparison conditions to FPB. Neither HRV nor cardiac contractility during PB15 significantly differed from baseline, ruling out potential influences due to visual pacing. During SPB6, the highest HRV and heart period values were recorded, replicating a large body of evidence (Laborde et al. 2022a). No significant change in cardiac contractility was found. A small decrease in PEP compared to baseline and PB15 may reflect higher preload mechanics rather than increased cardiac contractility due to the changes in intra-thoracic pressure.

4.1.2 | Fast-Paced Breathing

During 55 CPM, HRV decreased and cardiac contractility increased, consistent with the hypothesized reciprocally coupled response (Berntson et al. 1994). Descriptively, the FPB35 condition caused a milder reciprocal activation, however, only a reduction in HRV was statistically significant. Heart period did not significantly change during FPB35 nor FPB55. According to the tonic position, the decrease in RMSSD during FPB can be interpreted as reduced cardiac parasympathetic activity, with a more pronounced effect during FPB55 (Lehrer and Gevirtz 2014). Taking the phasic position, this result would instead signify decreased patterning of vagal effects upon heart rate during exhalation, but equal average cardiac vagal activity (Grossman and Taylor 2007). Nuanced interpretation of phasic autonomic responses is facilitated through changes in cardiac contractility. Due to low intra-thoracic movement, PEP is expected to be less influenced by cardiac preload and afterload changes in FPB than SPB, supporting its validity as an index of cardiac sympathetic (beta-adrenergic) activity (Lalande et al. 2012). The results show that the threshold for increasing cardiac contractility may lie between 35 and 55 CPM, providing initial evidence for a boundary condition beyond which reduced HRV is accompanied by elevated cardiac contractility.

The reciprocal nature of changes in RMSSD and PEP during FPB55 was not reflected in heart period values. Given that the mean heart period did not significantly decrease during FPB, one may find the phasic interpretation of vagal influences upon the heart more likely. However, such inferences have only been made in studies on paced breathing under pharmacological blockades (Grossman and Kollai 1993), and using heart period to infer changes in vagal influences under intact (non-blocked) conditions is not recommended according to current guidelines (Quigley et al. 2024). Inferences on phasic and tonic influences, therefore, cannot be transferred to the present study and remain speculative. To nonetheless attain a better understanding of the data, the spread and distribution of heart period values were assessed. During FPB35, the range of heart period values was highest among the conditions, suggesting diverse individual responses. In contrast, lower heart period values were more frequently observed during FPB55, although this trend was not reflected in the mean and did not reach statistical significance. Critically, decreased heart period in a subset of participants could

not be explained by differences in breathing depth, one of the influencing factors identified in the literature (Sroufe 1971; Tipton et al. 2017). To statistically confirm the descriptive trend found in our data, future studies require larger samples. Aside from the statistical considerations, the lack of a consistent decrease in heart period may have occurred due to mechanical factors. With small intrathoracic-pressure swings due to shallow breathing, mechanical stimulation is low (Angus et al. 2023). This may explain a more effective decrease in heart period during FPB with active forceful exhalation, commonly performed in Kapalbhathi practice (e.g., Nayak et al. 2015), compared to passive FPB in the present study.

Performance-facilitating effects of FPB have been previously found across a wide range of frequencies, raising the question of which cardiac changes accompany and contribute to motor-cognitive performance. Reported effects include reduced reaction and movement times during a fine motor skill task in a 25 CPM condition (Buchanan and Janelle 2021), enhanced perceived alertness, energy, and focus in a driving simulator while breathing at an individualized frequency (up to 46 CPM; Balters et al. 2018), and increased hand grip strength (Telles et al. 2014) in a 60 CPM condition. In light of our findings, the effects of the former study may be primarily accompanied by reduced HRV, while reciprocally coupled activation may play a more prominent role in the latter example. Systematic investigation of changes in motor-cognitive performance in response to FPB exercises is required to specify which functions and skills are facilitated through different cardiac patterns (Laborde et al. 2022c). For instance, reciprocally coupled activation may be particularly beneficial for tasks involving moderate to high motor demands.

4.2 | Duration of Breathing-Induced Effects

Physiological changes induced by breathing modifications are typically short-lived, making the duration of cardiac effects highly relevant. In the present study, the changes in RMSSD and PEP dissipated during the 2-min washout periods, suggesting a rapid return to baseline values (Laborde et al. 2018). Consideration of shorter time frames showed that after FPB55, PEP values did not increase immediately but rather gradually, returning to baseline 20–30 s after the cessation of paced breathing. While this effect could be attributed to breathing frequency variations, the lack of correlation between PEP values and breathing frequency makes this interpretation unlikely. The identified pattern of PEP changes aligns closely with the one found in Czarnek et al. (2021), despite using a different manipulation and duration (e.g., a 5-min rescue operation task in virtual reality). These results suggest that not only during but also shortly after performing FPB55, cardiac contractility is increased, carrying implications for study design and practical implementation of FPB exercises.

4.3 | Cardio-Respiratory Coupling During Paced Breathing

Our findings replicated the high synchronization level during SPB6 (Lehrer and Gevirtz 2014) and identified a decrease in phase locking with increasing respiratory frequency. Weak

cardio-respiratory coupling during FPB indicates that at higher breathing rates, heart rate cannot effectively synchronize with respiration. Several factors may contribute to this phenomenon. The sympathetic changes may be too slow to modulate heart rate at higher breathing frequencies (Kromenacker et al. 2018; Saul 1990). Moreover, the shortened respiratory cycle during FPB corresponds to smaller differences in the R-R interval between inspiration and expiration, attenuating the respiratory sinus arrhythmia (Yasuma and Hayano 2004). Lastly, shallow breathing diminishes the afferent feedback from pulmonary stretch receptors, further contributing to the desynchronization of the cardio-respiratory coupling (Noble and Hochman 2019). While baseline coupling strength has been positively associated with fitness level and recovery capacity (Abreu et al. 2023), the implications of the acute reduction in cardio-respiratory coupling remain to be clarified. Further empirical investigation is required to elucidate its predictive role in health- and performance-related outcomes.

4.4 | Breathing Discomfort

Participants experienced increased breathing discomfort as the breathing frequency deviated from the spontaneous range, reaching the highest discomfort level during FPB55. The findings are consistent with previously reported discomfort patterns in physically active samples (6–35 CPM, Denot-Ledunois et al. 1998), extending it to 55 CPM. Moderate discomfort levels reported during FPB55 suggest that participants found breathing exercises feasible to perform. Nonetheless, individual differences should be noted, and discomfort levels may differ across populations and experience levels. The lack of significant correlation between breathing discomfort and cardiac indices suggests that the subjective sensation of discomfort may arise from other sources, such as difficulty following the pacer or breathlessness sensation that may be enhanced during both SPB and FPB (Buchanan and Janelle 2022; Laborde et al. 2022b).

4.5 | Added Value and Implications

At the theoretical level, the present study provides empirical support for the speculated cardiac pattern during FPB, reciprocally coupled activation, at 55 but not at 35 CPM (Balters et al. 2018; Malhotra et al. 2022). Cardiac changes during FPB are therefore frequency-dependent. Given that the prior joint assessment of HRV and cardiac contractility during FPB has been limited to 25 CPM and PEP measurement may have been affected by methodological limitations arising from postural changes (Houtveen et al. 2005), the present findings offer a valuable contribution to the field. Seminal studies on PEP (e.g., Berntson et al. 1994) similarly showed that different stressors aligned in the direction of the response but varied in the extent to which reciprocal sympathetic activation was evoked. Whereas 55 CPM appears sufficiently high to elicit reciprocal coupling, with low between-subject variance in RMSSD, a more individualized approach may be required in identifying the threshold for increasing cardiac contractility.

Beyond autonomic regulation, paced breathing techniques have been conceptually implicated in cognitive and emotional regulation (e.g., Gerritsen and Band 2018). The development of

theoretical models on paced breathing critically depends on our knowledge of what changes in HRV and cardiac contractility signify. The challenge of integrating partly opposing positions on RSA as a measure of phasic vagal influences or cardiac vagal activity is not unique to the present study and has been recognized in the literature (e.g., Gerritsen and Band 2018; Quigley et al. 2024). Specifically, the proponents of the phasic position question the focus on the cardiac vagal system in the neurovisceral integration model proposed by Thayer et al. (2009), arguing that respiratory entrainment of neural processes represents the primary mechanism (Ritz 2024). To advance this debate, new systematic investigations of the interactions among cardiac vagal activity, vagus nerve discharge, and HRV when implementing FPB and SPB in regulatory tasks are necessary. More comprehensive quantification likely requires the use of singular and dual pharmacological blockades with selective β -blockers to address previous limitations (Fumagalli et al. 2020), or invasive procedures. Such endeavors may reveal whether modifying breathing frequency can assist with testing the theoretical predictions of the neurovisceral integration model and clarify the role of cardiac vagal afferents (Gerritsen and Band 2018; Sevoz-Couche and Laborde 2022).

The methodological implications of the present study focus on the association between HRV and cardiac contractility. The lack of a significant correlation between RMSSD and PEP, despite reciprocal patterns at the group level, suggests independence of phasic vagal influences and cardiac contractility at the individual level. Inclusion of both RMSSD and PEP may provide further insights into the complexity of phasic autonomic response patterns, considering both main effects and interactions (Giuliano et al. 2018). Analytically, dynamic changes at the beat-to-beat scale, considered in cardio-respiratory coupling, should be extended to inter-beat intervals and PEP to gain insights into the time-sensitive relationship between these indices. This approach addresses the existing bias towards overemphasizing the vagal contribution by offering more mechanistic precision than the critiqued construct of sympatho-vagal balance (Eckberg 1997; Quigley et al. 2024; Ritz 2024).

4.6 | Study Limitations

Despite providing insights into cardiac patterns during FPB, certain limitations of the present study should be noted.

1. The study did not assess cognitive or emotion regulation, therefore, the reference to the neurovisceral integration model primarily addresses the autonomic regulation at the cardiac level and serves as context for future theoretical work on regulatory effects of FPB.
2. The respiration signal was recorded using a respiration transducer. Precision could be enhanced through techniques such as respiratory inductance plethysmography.
3. Quantifying the amount of partial pressure of carbon dioxide in exhaled breath using capnography may facilitate discrimination of shallow fast breathing from hyperventilation. Due to the concerns regarding the validity of this measurement at very high breathing frequencies and

additional discomfort caused by the nasal cannula, we decided against using capnography and prioritized more natural, unrestricted breathing.

4. Despite our focus on cardiac indices, we acknowledge that cardio-respiratory interactions are part of a larger system, including the chemoreceptor and baroreceptor activity, as well as associated blood pressure changes (Dick et al. 2014). Specification of how the respiratory pump affects cardiac preload and afterload during paced breathing is required, given the sensitivity of PEP to these processes (Cacioppo et al. 1994). The aforementioned processes were outside of this study's scope but represent important targets for future research.
5. The sample consisted of physically active individuals, used to experiencing higher breathing frequencies. This limits the generalizability of our findings to other populations, particularly regarding the duration of FPB exercises and associated discomfort levels.
6. Despite carefully following the ICG data collection recommendation, several datasets were excluded due to movement artifacts, a challenge recognized in the literature (Sheikh et al. 2020). In the main analysis of PEP, the a priori power sample size requirement was met, however, the exploratory analyses were slightly under-powered and should thus be interpreted with caution.

5 | Conclusions

Cardiac response patterns during FPB at 55 CPM corresponded to reciprocal activation, reflected in reduced HRV and increased cardiac contractility, whereas 35 CPM primarily decreased HRV. These frequency-dependent changes illustrate distinct response patterns during FPB, carrying implications for advancing conceptual models of paced breathing. Future research on FPB should provide a fine-grained assessment of cardiac contractility thresholds and examine to what extent different cardiac patterns predict performance in domains where high psychophysiological activation is regarded as facilitative.

Author Contributions

Maša Iskra: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, software, validation, visualization, writing – original draft, writing – review and editing. **Sylvain Laborde:** conceptualization, methodology, resources, supervision, writing – review and editing. **Tasha Poppa:** data curation, software (sole developer of the PLV-based cardio-respiratory coherence analysis and SD-based ensemble ICG/PEP estimation methods), conceptual and methodological contributions during data analysis, writing – original draft (methods sections related to PLV and ICG/PEP), visualization, writing – review and editing. **Caterina Salvotti:** conceptualization, methodology, resources, validation, writing – review and editing. **Elisa Weinand:** data curation, investigation, writing – review and editing. **Markus Raab:** supervision, writing – review and editing. **Laura Voigt:** conceptualization, funding acquisition, methodology, supervision, writing – review and editing.

Acknowledgments

Authors would like to thank Jonathan Klatt for assisting with data collection and Dr. Andrea Brancaccio for the development of the breathing pacer used in this study.

Funding

The study was supported through the internal research funding of the German Sport University Cologne (M.I., L.V.), and by the Research Foundation—Flanders (FWO): G0E0520N (T.P.).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Preregistered hypotheses, methods, and analyses, as well as materials and processed data are available on the Open Science Framework at <https://doi.org/10.17605/OSF.IO/DV8EQ>.

References

- Abreu, R. M. D., B. Cairo, and A. Porta. 2023. "On the Significance of Estimating Cardiorespiratory Coupling Strength in Sports Medicine." *Frontiers in Network Physiology* 2: 1114733. <https://doi.org/10.3389/fnetp.2022.1114733>.
- Angus, S. A., J. L. Taylor, L. M. Mann, et al. 2023. "Attenuating Intrathoracic Pressure Swings Decreases Cardiac Output at Different Intensities of Exercise." *Journal of Physiology* 601, no. 21: 4807–4821.
- Árbol, J. R., P. Perakakis, A. Garrido, J. L. Mata, M. C. Fernández-Santaella, and J. Vila. 2017. "Mathematical Detection of Aortic Valve Opening (B Point) in Impedance Cardiography: A Comparison of Three Popular Algorithms." *Psychophysiology* 54, no. 3: 350–357. <https://doi.org/10.1111/psyp.12799>.
- Bae, D., J. J. Matthews, J. J. Chen, and L. Mah. 2021. "Increased Exhalation to Inhalation Ratio During Breathing Enhances High-Frequency Heart Rate Variability in Healthy Adults." *Psychophysiology* 58, no. 11: e13905. <https://doi.org/10.1111/psyp.13905>.
- Balters, S., E. L. Murnane, J. A. Landay, and P. E. Paredes. 2018. *Breath Booster!: Proceedings of the 12th EAI International Conference on Pervasive Computing Technologies for Healthcare*. PervasiveHealth '18: 12th EAI International Conference on Pervasive Computing Technologies for Healthcare, New York NY USA. <https://doi.org/10.1145/3240925.3240939>.
- Banzett, R. B., C. R. O'Donnell, T. E. Guilfoyle, et al. 2015. "Multidimensional Dyspnea Profile: An Instrument for Clinical and Laboratory Research." *European Respiratory Journal* 45, no. 6: 1681–1691. <https://doi.org/10.1183/09031936.00038914>.
- Bellissimo, G., E. Leslie, V. Maestas, and M. Zuhl. 2020. "The Effects of Fast and Slow Yoga Breathing on Cerebral and Central Hemodynamics." *International Journal of Yoga* 13, no. 3: 207–212. https://doi.org/10.4103/ijoy.IJOY_98_19.
- Berens, P. 2009. "CircStat: A MATLAB Toolbox for Circular Statistics." *Journal of Statistical Software* 31, no. 10: 1–21. <https://doi.org/10.18637/jss.v031.i10>.
- Berntson, G. G., J. T. Cacioppo, P. F. Binkley, B. N. Uchino, K. S. Quigley, and A. Fieldstone. 1994. "Autonomic Cardiac Control. III. Psychological Stress and Cardiac Response in Autonomic Space as Revealed by Pharmacological Blockades." *Psychophysiology* 31, no. 6: 599–608.
- Biopac Systems Inc. 2022. "BIOPAC MP36R [Apparatus and Software]." <https://www.biopac.com/product/mp36r-systems/>.

- Buchanan, T. L., and C. M. Janelle. 2021. "Fast Breathing Facilitates Reaction Time and Movement Time of a Memory-Guided Force Pulse." *Human Movement Science* 76: 102762. <https://doi.org/10.1016/j.humov.2021.102762>.
- Buchanan, T. L., and C. M. Janelle. 2022. "Emotions and Ensuing Motor Performance Are Altered by Regulating Breathing Frequency: Implications for Emotion Regulation and Sport Performance." *Frontiers in Psychology* 13: 963711. <https://doi.org/10.3389/fpsyg.2022.963711>.
- Busch, V., W. Magerl, U. Kern, J. Haas, G. Hajak, and P. Eichhammer. 2012. "The Effect of Deep and Slow Breathing on Pain Perception, Autonomic Activity, and Mood Processing—An Experimental Study." *Pain Medicine* 13, no. 2: 215–228. <https://doi.org/10.1111/j.1526-4637.2011.01243.x>.
- Cacioppo, J. T., G. G. Berntson, P. F. Binkley, K. S. Quigley, B. N. Uchino, and A. Fieldstone. 1994. "Autonomic Cardiac Control. II. Noninvasive Indices and Basal Response as Revealed by Autonomic Blockades." *Psychophysiology* 31, no. 6: 586–598. <https://doi.org/10.1111/j.1469-8986.1994.tb02351.x>.
- Choe, A. S., B. Tang, K. R. Smith, et al. 2021. "Phase-Locking of Resting-State Brain Networks With the Gastric Basal Electrical Rhythm." *PLoS One* 16, no. 1: e0244756. <https://doi.org/10.1371/journal.pone.0244756>.
- Czarnek, G., M. Richter, and P. Strojny. 2021. "Cardiac Sympathetic Activity During Recovery as an Indicator of Sympathetic Activity During Task Performance." *Psychophysiology* 58, no. 2: e13724. <https://doi.org/10.1111/psyp.13724>.
- Denot-Ledunois, S., G. Vardon, P. Perruchet, and J. Gallego. 1998. "Effects of Voluntary Changes in Breathing Frequency on Respiratory Comfort." *Biological Psychology* 49, no. 1–2: 71–82. [https://doi.org/10.1016/S0301-0511\(98\)00027-1](https://doi.org/10.1016/S0301-0511(98)00027-1).
- Dick, T. E., Y. H. Hsieh, R. R. Dhinra, et al. 2014. "Cardiorespiratory Coupling: Common Rhythms in Cardiac, Sympathetic, and Respiratory Activities." *Progress in Brain Research* 209: 191–205. <https://doi.org/10.1016/B978-0-444-63274-6.00010-2>.
- Eckberg, D. L. 1997. "Sympathovagal Balance: A Critical Appraisal." *Circulation* 96, no. 9: 3224–3232.
- Esco, M. R., and A. A. Flatt. 2014. "Ultra-Short-Term Heart Rate Variability Indexes at Rest and Post-Exercise in Athletes: Evaluating the Agreement With Accepted Recommendations." *Journal of Sports Science and Medicine* 13, no. 3: 535–541.
- Faul, F., E. Erdfelder, A. Buchner, and A. G. Lang. 2009. "Statistical Power Analyses Using G* Power 3.1: Tests for Correlation and Regression Analyses." *Behavior Research Methods* 41, no. 4: 1149–1160.
- Fisher, J. P., C. N. Young, and P. J. Fadel. 2015. "Autonomic Adjustments to Exercise in Humans." *Comprehensive Physiology* 5, no. 2: 475–512. <https://doi.org/10.1002/j.2040-4603.2015.tb00614.x>.
- Fumagalli, C., N. Maurizi, N. Marchionni, and D. Fornasari. 2020. "β-Blockers: Their New Life From Hypertension to Cancer and Migraine." *Pharmacological Research* 151: 104587.
- Gerritsen, R. J., and G. P. Band. 2018. "Breath of Life: The Respiratory Vagal Stimulation Model of Contemplative Activity." *Frontiers in Human Neuroscience* 12: 397.
- Gersten, A., O. Gersten, A. Ronen, and Y. Cassuto. 1999. "The RR Interval Spectrum, the ECG Signal and Aliasing." In *Fourier Transforms—Century of Digitalization and Increasing Expectations*, 1–17. IntechOpen.
- Gholamrezaei, A., I. Van Diest, Q. Aziz, J. W. Vlaeyen, and L. Van Oudenhove. 2021. "Psychophysiological Responses to Various Slow, Deep Breathing Techniques." *Psychophysiology* 58, no. 2: e13712. <https://doi.org/10.1111/psyp.13712>.
- Gilbert, C. 1999. "Breathing and the Cardiovascular System." *Journal of Bodywork and Movement Therapies* 3, no. 4: 215–224. [https://doi.org/10.1016/S1360-8592\(99\)80006-0](https://doi.org/10.1016/S1360-8592(99)80006-0).
- Giuliano, R. J., C. M. Karns, T. A. Bell, et al. 2018. "Parasympathetic and Sympathetic Activity Are Associated With Individual Differences in Neural Indices of Selective Attention in Adults." *Psychophysiology* 55, no. 8: e13079. <https://doi.org/10.1111/psyp.13079>.
- Grossman, P. 2023. "Fundamental Challenges and Likely Refutations of the Five Basic Premises of the Polyvagal Theory." *Biological Psychology* 180: 108589.
- Grossman, P., J. Karemaker, and W. Wieling. 1991. "Prediction of Tonic Parasympathetic Cardiac Control Using Respiratory Sinus Arrhythmia: The Need for Respiratory Control." *Psychophysiology* 28, no. 2: 201–216.
- Grossman, P., and M. Kollai. 1993. "Respiratory Sinus Arrhythmia, Cardiac Vagal Tone, and Respiration: Within-and Between-Individual Relations." *Psychophysiology* 30, no. 5: 486–495.
- Grossman, P., and E. W. Taylor. 2007. "Toward Understanding Respiratory Sinus Arrhythmia: Relations to Cardiac Vagal Tone, Evolution and Biobehavioral Functions." *Biological Psychology* 74, no. 2: 263–285.
- Herrero, J. L., S. Khuvis, E. Yeagle, M. Cerf, and A. D. Mehta. 2018. "Breathing Above the Brain Stem: Volitional Control and Attentional Modulation in Humans." *Journal of Neurophysiology* 119: 145–159. <https://doi.org/10.1152/jn.00551.2017>.
- Hill, L. K., A. Siebenbrock, J. J. Sollers, and J. F. Thayer. 2009. "Are All Measures Created Equal? Heart Rate Variability and Respiration." *Biomedical Sciences Instrumentation* 45: 71–76.
- Houtveen, J. H., P. F. Groot, and E. J. De Geus. 2005. "Effects of Variation in Posture and Respiration on RSA and Pre-Ejection Period." *Psychophysiology* 42, no. 6: 713–719. <https://doi.org/10.1111/j.1469-8986.2005.00363.x>.
- Huang, C., R. N. Gervitz, J. Onton, and J. R. Criado. 2018. "Investigation of Vagal Afferent Functioning Using the Heartbeat Event Related Potential." *International Journal of Psychophysiology* 131: 113–123.
- JASP Team. 2024. "JASP (Version 0.18.3)[Computer software]."
- Joshi, M., and S. Telles. 2009. "A Nonrandomized Non-Naive Comparative Study of the Effects of Kapalabhati and Breath Awareness on Event-Related Potentials in Trained Yoga Practitioners." *Journal of Alternative and Complementary Medicine* 15, no. 3: 281–285. <https://doi.org/10.1089/acm.2008.0250>.
- Karlocai, K., G. Jokkel, and M. Kollai. 1998. "Changes in Left Ventricular Contractility With the Phase of Respiration." *Journal of the Autonomic Nervous System* 73, no. 2–3: 86–92.
- Klaperski, S., B. von Dawans, M. Heinrichs, and R. Fuchs. 2013. "Does the Level of Physical Exercise Affect Physiological and Psychological Responses to Psychosocial Stress in Women?" *Psychology of Sport and Exercise* 14, no. 2: 266–274. <https://doi.org/10.1016/j.psychsport.2012.11.003>.
- Kromenacker, B. W., A. A. Sanova, F. I. Marcus, J. J. Allen, and R. D. Lane. 2018. "Vagal Mediation of Low-Frequency Heart Rate Variability During Slow Yogic Breathing." *Psychosomatic Medicine* 80, no. 6: 581–587. <https://doi.org/10.1097/PSY.0000000000000603>.
- Laborde, S., M. S. Allen, U. Borges, et al. 2022a. "Effects of Voluntary Slow Breathing on Heart Rate and Heart Rate Variability: A Systematic Review and a Meta-Analysis." *Neuroscience & Biobehavioral Reviews* 138: 104711. <https://doi.org/10.1016/j.neubiorev.2022.104711>.
- Laborde, S., M. S. Allen, U. Borges, et al. 2022b. "Psychophysiological Effects of Slow-Paced Breathing at Six Cycles Per Minute With or Without Heart Rate Variability Biofeedback." *Psychophysiology* 59, no. 1: e13952. <https://doi.org/10.1111/psyp.13952>.
- Laborde, S., E. Mosley, and A. Mertgen. 2018. "Vagal Tank Theory: The Three Rs of Cardiac Vagal Control Functioning—Resting, Reactivity,

- and Recovery." *Frontiers in Neuroscience* 12: 458. <https://doi.org/10.3389/fnins.2018.00458>.
- Laborde, S., E. Mosley, and J. F. Thayer. 2017. "Heart Rate Variability and Cardiac Vagal Tone in Psychophysiological Research—Recommendations for Experiment Planning, Data Analysis, and Data Reporting." *Frontiers in Psychology* 8: 213. <https://doi.org/10.3389/fpsyg.2017.00213>.
- Laborde, S., N. Zammit, M. Iskra, et al. 2022c. "The Influence of Breathing Techniques on Physical Sport Performance: A Systematic Review and Meta-Analysis." *International Review of Sport and Exercise Psychology* 17, no. 2: 1222–1277. <https://doi.org/10.1080/1750984X.2022.2145573>.
- Lachaux, J. P., E. Rodriguez, J. Martinerie, and F. J. Varela. 1999. "Measuring Phase Synchrony in Brain Signals." *Human Brain Mapping* 8, no. 4: 194–208. [https://doi.org/10.1002/\(SICI\)1097-0193\(1999\)8:4<194::AID-HBM4>3.0.CO;2-C](https://doi.org/10.1002/(SICI)1097-0193(1999)8:4<194::AID-HBM4>3.0.CO;2-C).
- Lalande, S., C. E. Luoma, A. D. Miller, and B. D. Johnson. 2012. "Effect of Changes in Intrathoracic Pressure on Cardiac Function at Rest and During Moderate Exercise in Health and Heart Failure." *Experimental Physiology* 97, no. 2: 248–256.
- Lehrer, P. M., and R. Gevirtz. 2014. "Heart Rate Variability Biofeedback: How and Why Does It Work?" *Frontiers in Psychology* 5: 756. <https://doi.org/10.3389/fpsyg.2014.00756>.
- Lenneman, J. K., and R. W. Backs. 2009. "Cardiac Autonomic Control During Simulated Driving With a Concurrent Verbal Working Memory Task." *Human Factors* 51, no. 3: 404–418. <https://doi.org/10.1177/0018720809337716>.
- Lent, M. C., K. J. Perry, G. R. Perhamus, C. Buck, D. Murray-Close, and J. M. Ostrov. 2024. "Is Autonomic Functioning Distinctly Associated With Anxiety and Unsociability in Preschoolers?" *International Journal of Psychophysiology* 200: 112343. <https://doi.org/10.1016/j.ijpsycho.2024.112343>.
- Ley, R. 1994. "Breathing and the Psychology of Emotion, Cognition, and Behavior." In *Behavioral and Psychological Approaches to Breathing Disorders*, 81–95. Springer US.
- Malhotra, V., D. Javed, S. Wakode, R. Bharshankar, N. Soni, and P. K. Porter. 2022. "Study of Immediate Neurological and Autonomic Changes During Kapalbhathi Pranayama in Yoga Practitioners." *Journal of Family Medicine and Primary Care* 11, no. 2: 720–727. https://doi.org/10.4103/jfmpc.jfmpc_1662_21.
- McArdle, W. D., F. I. Katch, and V. L. Katch. 2015. *Exercise Physiology: Nutrition, Energy, and Human Performance*. Lippincott Williams & Wilkins.
- Nalbandian, M., Z. Radak, J. Taniguchi, and T. Masaki. 2017. "How Different Respiratory Rate Patterns Affect Cardiorespiratory Variables and Performance." *International Journal of Exercise Science* 10, no. 3: 322. <https://doi.org/10.70252/WSFL7054>.
- Nayak, R., S. Prakash, R. K. Yadav, and K. Upadhyay-Dhungel. 2015. "Kapalbhathi Changes Cardiovascular Parameters." *Janaki Medical College Journal of Medical Science* 3, no. 2: 43–49. <https://doi.org/10.3126/jmcjms.v3i2.16080>.
- Newlin, D. B., and R. W. Levenson. 1979. "Pre-Ejection Period: Measuring Beta-Adrenergic Influences Upon the Heart." *Psychophysiology* 16, no. 6: 546–552.
- Nigg, J. T. 2017. "Annual Research Review: On the Relations Among Self-Regulation, Self-Control, Executive Functioning, Effortful Control, Cognitive Control, Impulsivity, Risk-Taking, and Inhibition for Developmental Psychopathology." *Journal of Child Psychology and Psychiatry* 58, no. 4: 361–383.
- Noble, D. J., and S. Hochman. 2019. "Hypothesis: Pulmonary Afferent Activity Patterns During Slow, Deep Breathing Contribute to the Neural Induction of Physiological Relaxation." *Frontiers in Physiology* 10: 1176. <https://doi.org/10.3389/fphys.2019.01176>.
- Peirce, J. W., R. J. Hirst, and M. R. MacAskill. 2022. *Building Experiments in PsychoPy*. 2nd ed. Sage.
- Peng, C. K., I. C. Henry, J. E. Mietus, et al. 2004. "Heart Rate Dynamics During Three Forms of Meditation." *International Journal of Cardiology* 95, no. 1: 19–27. <https://doi.org/10.1016/j.ijcard.2003.02.006>.
- Penttilä, J., A. Helminen, T. Jartti, et al. 2001. "Time Domain, Geometrical and Frequency Domain Analysis of Cardiac Vagal Outflow: Effects of Various Respiratory Patterns." *Clinical Physiology* 21, no. 3: 365–376. <https://doi.org/10.1046/j.1365-2281.2001.00337.x>.
- Pinna, G. D., R. Maestri, M. T. La Rovere, E. Gobbi, and F. Fanfulla. 2006. "Effect of Paced Breathing on Ventilatory and Cardiovascular Variability Parameters During Short-Term Investigations of Autonomic Function." *American Journal of Physiology. Heart and Circulatory Physiology* 290, no. 1: H424–H433. <https://doi.org/10.1152/ajpheart.00438.2005>.
- Pradhan, B. 2013. "Effect of Kapalabhati on Performance of Six-Letter Cancellation and Digit Letter Substitution Task in Adults." *International Journal of Yoga* 6, no. 2: 128. <https://doi.org/10.4103/0973-6131.113415>.
- Quigley, K. S., P. J. Gianaros, G. J. Norman, J. R. Jennings, G. G. Berntson, and E. J. de Geus. 2024. "Publication Guidelines for Human Heart Rate and Heart Rate Variability Studies in Psychophysiology—Part 1: Physiological Underpinnings and Foundations of Measurement." *Psychophysiology* 61, no. 9: e14604.
- R Core Team. 2022. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.
- Ritz, T. 2024. "Putting Back Respiration Into Respiratory Sinus Arrhythmia or High-Frequency Heart Rate Variability: Implications for Interpretation, Respiratory Rhythmicity, and Health." *Biological Psychology* 185: 108728.
- Saul, J. P. 1990. "Beat-To-Beat Variations of Heart Rate Reflect Modulation of Cardiac Autonomic Outflow." *Physiology* 5, no. 1: 32–37.
- Sevoz-Couche, C., and S. Laborde. 2022. "Heart Rate Variability and Slow-Paced Breathing: When Coherence Meets Resonance." *Neuroscience & Biobehavioral Reviews* 135: 104576. <https://doi.org/10.1016/j.neubiorev.2022.104576>.
- Shaffer, F., and Z. M. Meehan. 2020. "A Practical Guide to Resonance Frequency Assessment for Heart Rate Variability Biofeedback." *Frontiers in Neuroscience* 14: 570400.
- Sharma, V. K., M. Rajajeyakumar, S. Velkumary, et al. 2014. "Effect of Fast and Slow Pranayama Practice on Cognitive Functions in Healthy Volunteers." *Journal of Clinical and Diagnostic Research: JCDR* 8, no. 1: 10–13. <https://doi.org/10.7860/JCDR/2014/7256.3668>.
- Sheikh, S. A. A., A. Shah, O. Levantsevych, et al. 2020. "An Open-Source Automated Algorithm for Removal of Noisy Beats for Accurate Impedance Cardiogram Analysis." *Physiological Measurement* 41, no. 7: 075002.
- Smith, R., J. F. Thayer, S. S. Khalsa, and R. D. Lane. 2017. "The Hierarchical Basis of Neurovisceral Integration." *Neuroscience & Biobehavioral Reviews* 75: 274–296. <https://doi.org/10.1016/j.neubiorev.2017.02.003>.
- Sonetti, D. A., T. J. Wetter, D. F. Pegelow, and J. A. Dempsey. 2001. "Effects of Respiratory Muscle Training Versus Placebo on Endurance Exercise Performance." *Respiration Physiology* 127, no. 2–3: 185–199. [https://doi.org/10.1016/S0034-5687\(01\)00250-X](https://doi.org/10.1016/S0034-5687(01)00250-X).
- Sroufe, L. A. 1971. "Effects of Depth and Rate of Breathing on Heart Rate and Heart Rate Variability." *Psychophysiology* 8, no. 5: 648–655. <https://doi.org/10.1111/j.1469-8986.1971.tb00500.x>.
- Telles, S., S. K. Sharma, A. Yadav, N. Singh, and A. Balkrishna. 2014. "Immediate Changes in Muscle Strength and Motor Speed Following Yoga Breathing." *Indian Journal of Physiology and Pharmacology* 58, no. 1: 22–29.

Thayer, J. F., A. L. Hansen, E. Saus-Rose, and B. H. Johnsen. 2009. "Heart Rate Variability, Prefrontal Neural Function, and Cognitive Performance: The Neurovisceral Integration Perspective on Self-Regulation, Adaptation, and Health." *Annals of Behavioral Medicine* 37, no. 2: 141–153.

Tipton, M. J., A. Harper, J. F. Paton, and J. T. Costello. 2017. "The Human Ventilatory Response to Stress: Rate or Depth?" *Journal of Physiology* 595, no. 17: 5729–5752.

Yasuma, F., and J. I. Hayano. 2004. "Respiratory Sinus Arrhythmia: Why Does the Heartbeat Synchronize With Respiratory Rhythm?" *Chest* 125, no. 2: 683–690.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Characteristics of the respiratory and inter-beat-interval (IBI) signals in power spectrum across breathing conditions ($M \pm SD$). **Figure S1:** Examples of intra-individual cardio-respiratory coupling across the breathing conditions. **Table S2:** Descriptive statistics for phase lock values (PLVs) and mean phase angle differences ($M \pm SD$).